



Introduction to Knowledge Graphs

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Before we start...

Contact

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 - `ernesto.jimenez-ruiz@citystgeorges.ac.uk`
 - `https://ernestojimenezruiz.github.io/`

GitHub Repository

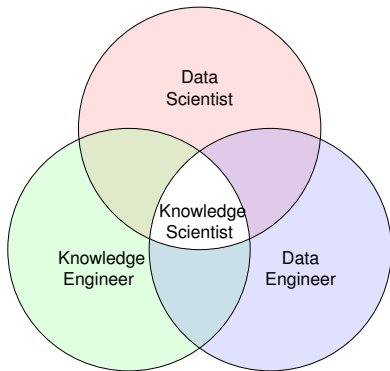
`https://github.com/city-knowledge-graphs/kg-course-valgrai`

Course Organization

1. [Introduction to Knowledge Graphs](#)
2. Modelling Ontologies with OWL (semantics and reasoning)
3. From Tabular Data to Knowledge Graphs
4. Querying Knowledge Graphs and Graph Database Solutions
5. Knowledge Graphs and Language Models

Skills to be gained

- **Data Engineer:** harnesses and collects data.
- **Data Scientist:** draws value from data.
- **Knowledge Engineer:** encodes domain expertise.
- **Knowledge Scientist:** adds context to the data to make it more useful, clean, reliable and ready to be used.



George Fletcher and others. Knowledge Scientists: Unlocking the data-driven organization. 2020

Introduction

The Vision of a Semantic Web: Tim Berners-Lee

«I have a dream for the Web [in which computers] become capable of analyzing all the data on the Web—the content, links, and transactions between people and computers. A ‘Semantic Web’, which should make this possible, has yet to emerge, but when it does, the day-to-day mechanisms of trade, bureaucracy and our daily lives will be handled by machines talking to machines. The ‘intelligent agents’ people have touted for ages will finally materialize.»



Quoted from: *Weaving the Web: The Original Design and Ultimate Destiny of the World Wide Web*. Tim Berners-Lee with Mark Fischetti. Harper San Francisco, 1999.

About Semantic Web technologies

- Originally designed as an extension of the World Wide Web to **make Internet data machine-readable**.
- Nowadays also covers the encoding of the semantics of arbitrary data (*e.g.*, enterprise data): **Knowledge Graphs**

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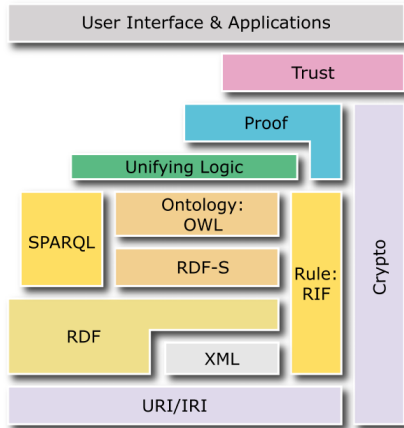
(*) How LLMs fit within the Semantic Web / Knowledge Graphs vision?

Semantic Web technology stack (i)

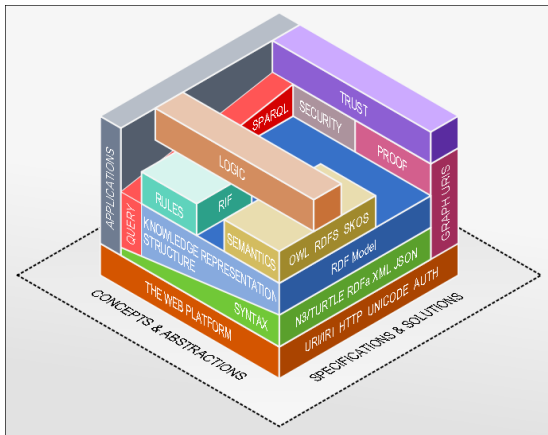
- The **World Wide Web Consortium (W3C)** is an international community that develops open standards to ensure the long-term growth of the Web: <https://www.w3.org/>
- **Why standards?**
 - broader industry (and academic) agreement,
 - interoperability across organizations and applications,
 - avoids vendor lock-in of a particular (exchange) format.
- The Semantic Web (as the Web) is built around W3C standards: the **Semantic Web stack**.

Semantic Web technology stack (ii)

- Identification: **URI/IRI**
- Data Representation: **RDF**
- Knowledge and Reasoning: **RDFS and OWL**
- Query language: **SPARQL**
- Exchange format: **XML**



Semantic Web technology stack (iii)



Overview standards. <https://csiro-enviro-informatics.github.io/info-engineering/standards.html>.

Image created by Benjamin Nowack (CC BY 3.0 Deed).

Challenges and Future of the Semantic Web (i)

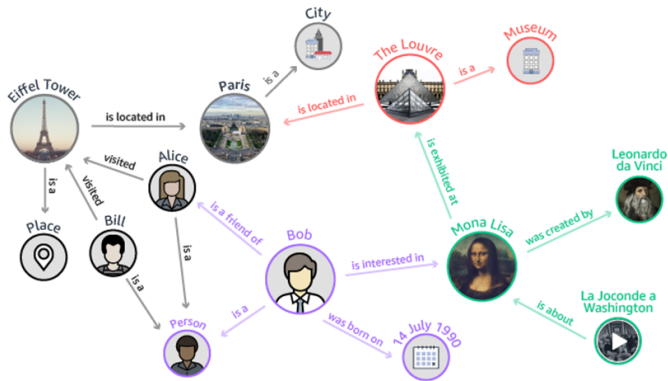
- Distributed **Web of Data** applied to the whole Web is **challenging**
 - Data should be provided in RDF
 - Data should be linked to knowledge (*e.g.*, about cinemas, movies, genres, places, etc.)
 - Data providers should agree about the knowledge in intersecting domains
 - Consistency and trust of the data and knowledge
 - Smart agents in the Web

Challenges and Future of the Semantic Web (ii)

- **We are moving forward:**
 - There was also scepticism about the Web
 - **Examples:** Wikidata, DBPedia, the Linked Open Data Cloud, Bio2RDF, Google's Knowledge Graph.
 - Semantic Web within an organisation: **Graph(s) of Knowledge**
 - Combination of LLMs and KGs.

Knowledge Graphs

- **(data graph)** capture information about entities of interest (like people, places or events), and
- **(knowledge)** with an underlying formal model (ontology)



Knowledge Graphs: <https://arxiv.org/abs/2003.02320>.

Graph(s) of Knowledge (i)

- Semantic Web in more **controlled scenarios**
- Not new but fresh notion of **knowledge graph in industry**
- Availability of (mature) **Semantic Web technology**
- Enterprise data as a knowledge graph to **drive products** and make them more “**intelligent**”

Keynote by **Ora Lassila** (Principal Technologist, Amazon Neptune) at ISWC 2024. **Semantic Web and AI: Can we finally realize the vision?** https://videlectures.net/videos/iswc2024_lassila_web_and_ai.

Graph(s) of Knowledge (ii)

- Identify and integrate disparate resources in the Web (*e.g.*, transport, cinemas, films)
- Integrate data within an organisation (*e.g.*, multiple data sources and departments)
- Combine life science data from genetic, pharmaceutical, patient database, etc.
- Cross-reference disparate digital libraries

Graph(s) of Knowledge (iii)

- In data science and AI, knowledge graphs are commonly used to:
 - Serve as bridges between humans and systems;
 - Facilitate access to and integration of data sources;
 - Represent enterprise data to drive products and make them more “intelligent”; and
 - Add context and depth to other, more data-driven AI techniques such as machine learning (*e.g.*, **Neurosymbolic AI**)

Interest Group @ The Alan Turing Institute: <https://www.turing.ac.uk/research/interest-groups/knowledge-graphs>

Ontologies (the "K" in the KG)

- “**Formal** specifications of a shared domain conceptualization”
- “Abstract symbolic representations of a domain expressed in a **formal** language”

Thomas R. Gruber. Towards Principles for the Design of Ontologies Used for Knowledge Sharing. 1993
Pim Borst, Hans Akkermans, and Jan Top. Engineering ontologies. 1999.

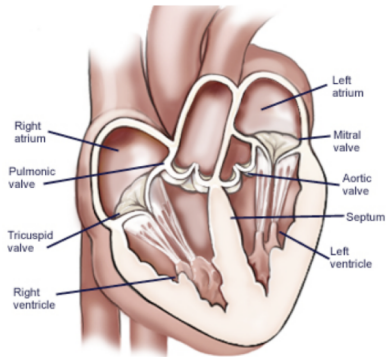
Ontologies as domain models (i)

- A model is a simplified (abstract) representation of certain aspects of the real world.
- Models help people communicate.
- Models explain and make predictions.
- Models mediate among multiple viewpoints.

Dean Allemang, James Hendler. Semantic Web for the Working Ontologist: Effective Modeling in RDFS and OWL.

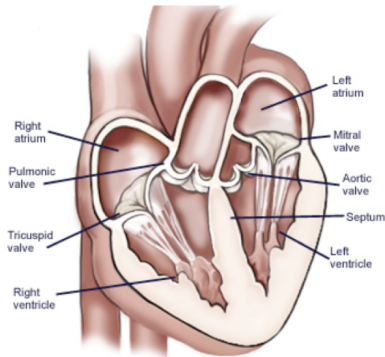
Ontologies as domain models (ii)

- include **vocabulary** relevant to a domain
- specify meaning (**semantics**) of terms
 - Heart is a muscular organ that is part of the circulatory system



Ontologies as domain models (ii)

- include **vocabulary** relevant to a domain
- specify meaning (**semantics**) of terms
 - Heart is a muscular organ that is part of the circulatory system
- are **formalised** using a suitable logic language
 - $\forall x.[Heart(x) \rightarrow$
 $MuscularOrgan(x) \wedge$
 $\exists y.[isPartOf(x, y) \wedge$
 $CirculatorySystem(y)]]$



Why Ontologies and KGs? (Experiences from Industry)

- Knowledge graphs are **not complex**: they are the simplest way to layer human expertise onto stored data.

Juan Sequeda and Ora Lassila. **Designing and Building Enterprise Knowledge Graphs**. 2021

Ora Lassila. **On the broad applicability of Semantic Web technologies**. (COST DKG Talk Series. 2021

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- **Unifying logical representation** for data and semantics to get out the data science enterprise mess.

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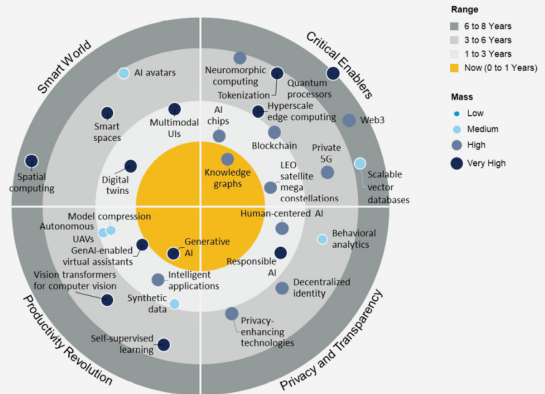
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Knowledge Graphs according to Gartner (i)

30 Emerging Technologies That Will Guide Your Business Decisions:
Gartner Web

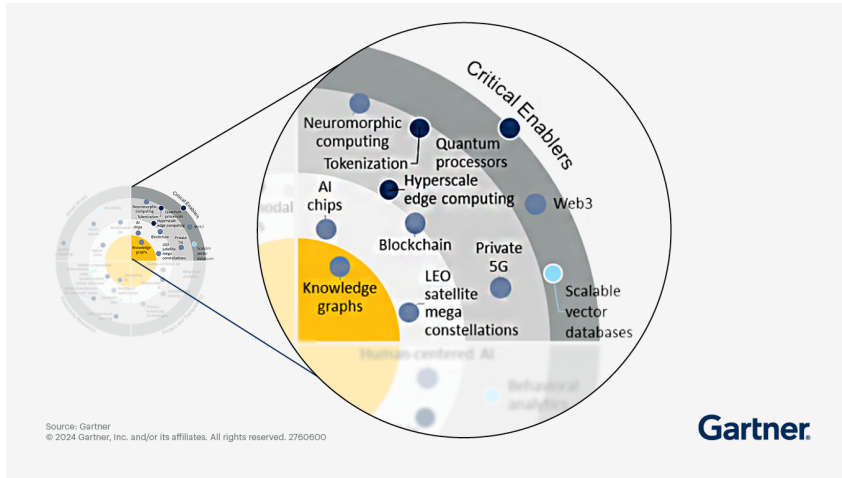
Impact Radar for 2024



Source: Gartner
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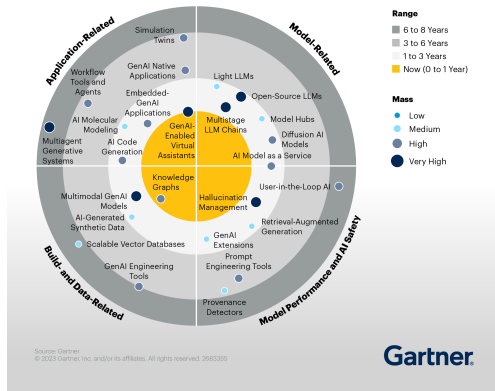
Gartner.

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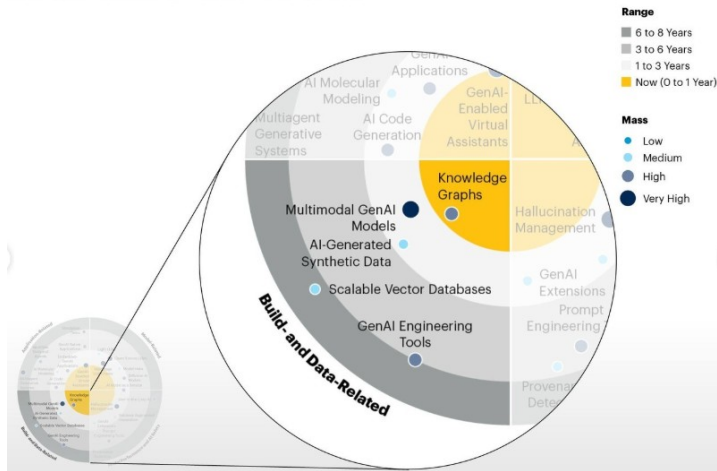


Knowledge Graphs according to Gartner (iii)

Impact Radar for Generative AI



Knowledge Graphs according to Gartner (iv)



RDF-based Knowledge Graphs

Graph Data Models

Data Graphs

- **Foundation** of any Knowledge Graph
- **Intuitive abstractions** to capture entities and their relationships.
- **Do not require schema** to start adding data.
- **Flexibility** to assemble data from multiple sources.

Data Graphs

- **Foundation** of any Knowledge Graph
- **Intuitive abstractions** to capture entities and their relationships.
- **Do not require schema** to start adding data.
- **Flexibility** to assemble data from multiple sources.
- Enable **queries over paths** of arbitrary length.
- Allow the use of **graph analytics** techniques.
- (Potential) correspondence with **logical** unary and binary **predicates**.

Types of graph-structured data models (i)

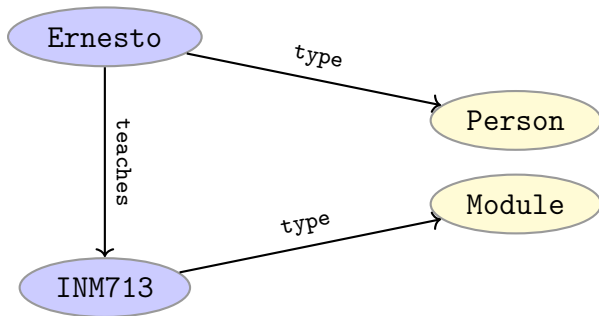
- Directed edge-labelled (multi)graphs (multi-relational graphs)
- Property graphs
- Heterogeneous graphs (heterogeneous information within a node)
- Hypergraphs (edges connecting set of nodes)
- Hypernodes (nested graphs in a node)

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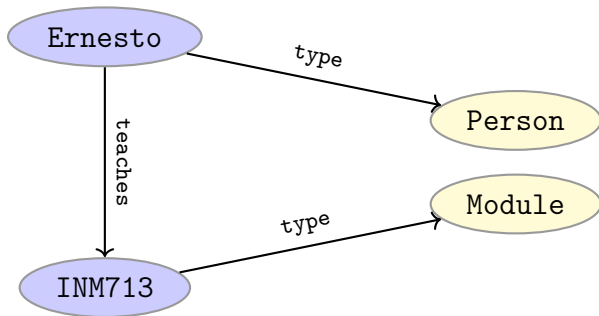
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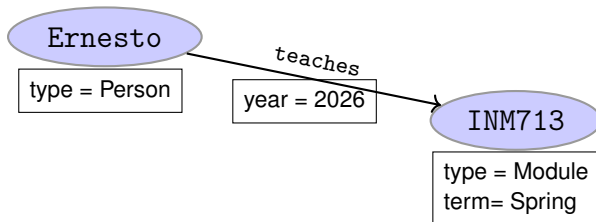
Types of graph-structured data models (ii)

Directed edge-labelled (multi)graphs. [\(In this course\).](#)



Types of graph-structured data models (iii)

Property graphs



RDF: Resource Description Framework

RDF in a nutshell

- Resource Description Framework (RDF)
- A **standardised data model** based on the **directed edge-labelled graph** model.
 - **Nodes**: Internationalised Resource Identifiers (IRIs), literals, and blank nodes (nodes without identifier).
 - **Edges**: IRIs
- **W3C recommendation.**
- Conceptual **modelling of resources.**

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 - **Edges**: IRIs
- **W3C recommendation.**
- Conceptual **modelling of resources.**
- In this course we will study **RDF-based Knowledge Graphs.**

All you need are triples

RDF Triples

- All information in RDF is expressed using a *triple* pattern.
- A triple consists of a **subject**, a **predicate**, and an **object**.

Examples:

subject	predicate	object
England	has capital	London
England	has king	Charles III
Charles III	born year	1948

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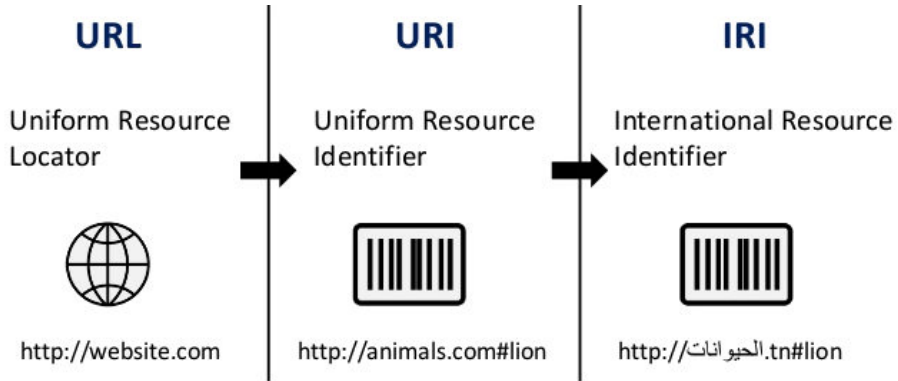
Examples:

subject	predicate	object
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- Another word for an RDF triple is a *statement* or *fact*.
- The elements of an RDF triple are either: *URI references*, *literals*, *blank nodes*.

All you need are triples... and URIs

Identifying resources



Identifying resources in RDF

- RDF talks about *resources*.
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- E.g., in `dbpedia.org` KG graph:

England: `http://dbpedia.org/resource/England`

has capital: `http://dbpedia.org/ontology/capital`

London: `http://dbpedia.org/resource/London`

has king: `http://dbpedia.org/ontology/monarch`

Charles III: `http://dbpedia.org/resource/Charles,\_Prince\_of\_Wales`

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- URIs/IRIs are not necessarily dereferenceable; but they could be: *Web of Data*.

Building an URI

scheme: [//authority]path[?query] [#fragment] (in brackets optional elements)

- scheme: http, https, ftp, file, etc.
- authority: typically the domain, //www.city.ac.uk
- path: /sst/cs/academics
- query: additional information for a Web service (not used in our URIs)
- fragment: to address a sub-part of a retrieved resource, #ernesto

e.g., <http://www.city.ac.uk/sst/cs/academics#ernesto>

Pascal Hitzler, Markus Krotzsch, Sebastian Rudolph. Foundations of Semantic Web Technologies. CRC Press 2009.

Identifying resources: URIs and (local) CURIs

- **URIs are often long** and hard to read and write.
- Most serialisations use an **abbreviation** mechanism.
 - Define “prefixes”, “namespaces”.
 - RDF/XML format: XML namespaces and entities.

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 - Define “prefixes”, “namespaces”.
 - RDF/XML format: XML namespaces and entities.
- E.g., in Turtle serialisation:

```
@prefix dbr: <http://dbpedia.org/resource/> .  
@prefix dbo: <http://dbpedia.org/ontology/> .
```
- A *CURI* (Compact URI) or *QName* like ‘dbr:London’ stands for `http://dbpedia.org/resource/London`

URIs, QNames and data value

- We can then state that England's capital is London as:

```
<http://dbpedia.org/resource/England  
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- Or using prefixes:

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```

- But what if we want to state that London's population is **9,304,000**?
- We cannot have one URI for every integer, decimal number, string, etc.

All you need are triples... and URIs... and literals

Literals in RDF

- Literals are used to represent **data values**.
- A literal in a RDF consist of these elements:
 - A lexical form: “London”, “9,304,000”, “Londres”
 - A (supported) datatype IRI (from OWL, RDF or XML Schema datatypes): *e.g.*, `^^xsd:integer`
 - A language tag (*e.g.*, “en”, “es”)

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 - A language tag (*e.g.*, “en”, “es”)
- Examples:
 - `dbr:London dbo:population "9,304,000"^^xsd:integer .`
 - `dbr:London rdfs:label "London"^^xsd:string .`
 - `dbr:London rdfs:label "Londres"@es .`

RDF Graph = set of triples with URIs and literals (and blank nodes)

Example RDF-based (Knowledge) Graph

London is a city in England called Londres in Spanish

```
dbr:london rdf:type dbo:City .
```

```
dbr:london dbo:locationCountry dbr:england .
```

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```

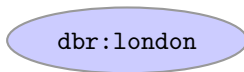
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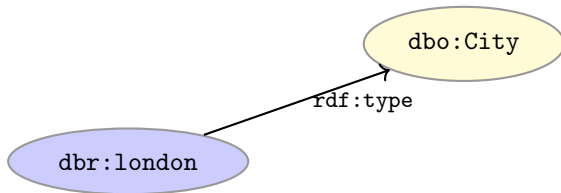
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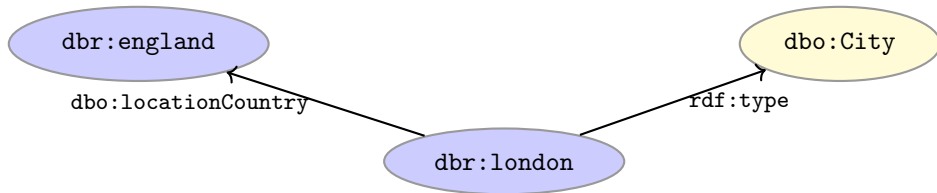
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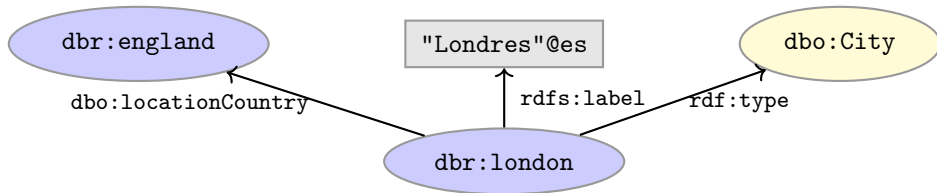
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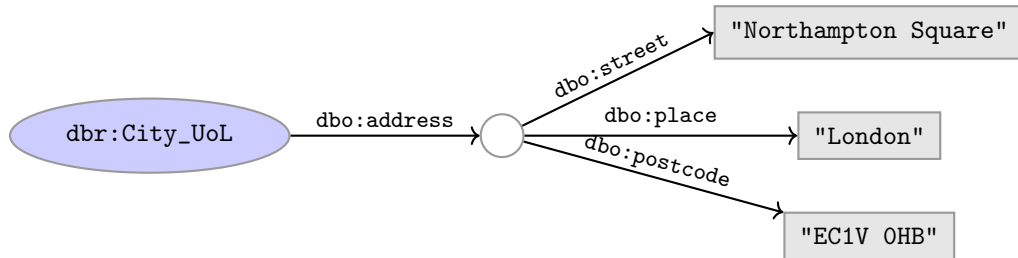


RDF Named Graphs

- We can **refer to a set of triples** with a name (*i.e.*, URI)
- Why?
 - **Independent** sources: 1 file, 1 graph.
 - Add specific **context** to a set of triples (*e.g.*, provenance).
 - Allowing to **query** a specific set of triples.
- Triples in a named graph sometimes referred as **quads**.

RDF Graphs: Blank nodes

- Blank nodes are like resources without a URI.
- Use when resource is unknown, or has no (natural) identifier. *e.g.*,



RDF Triple Grammar

– Literals and blank nodes may not appear everywhere in triples:

- | | s | p | o |
|---|---|---|---|
| • URI references may occur in all positions | ✓ | ✓ | ✓ |
| • Literals may only occur in object position | ✗ | ✗ | ✓ |
| • Blank nodes can not occur in predicate position | ✓ | ✗ | ✓ |

RDF Triple Grammar

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	s	p	o
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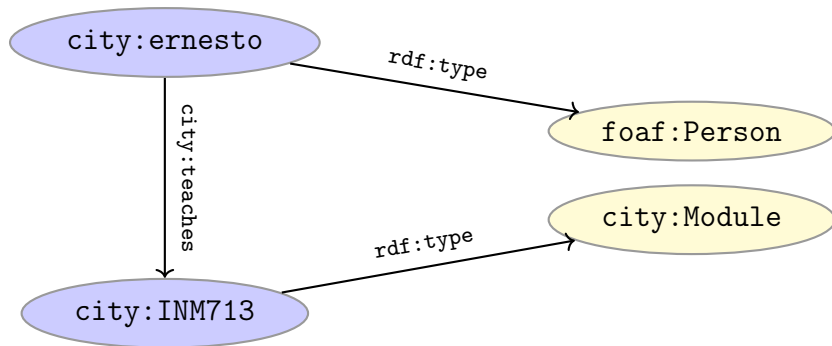
- Why?
 - Literals are just values, no relationships from literals allowed.
 - Blank nodes in predicate position deemed “too meaningless” and confusing.

RDF and Property Graphs

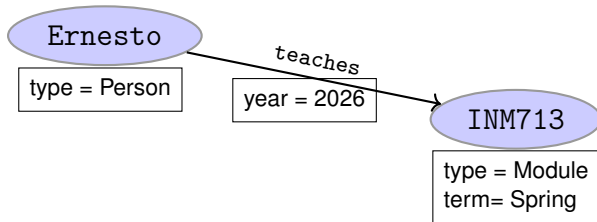
Annotating statements/triples in RDF?

How to encode that “**Ernesto teaches INM713 in 2026**”?

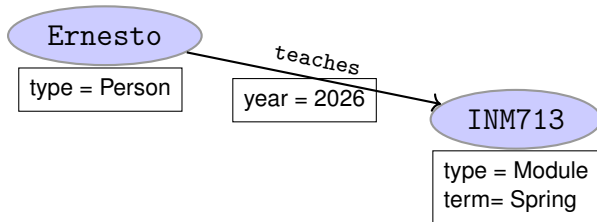
(i.e., **Metadata of a triple**: provenance, beliefs, uncertainty, creator, time, etc.)



Using property Graphs

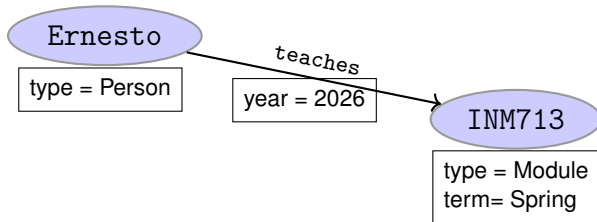


Using property Graphs



But can we do it in RDF?

Using property Graphs



But can we do it in RDF? **Yes!**

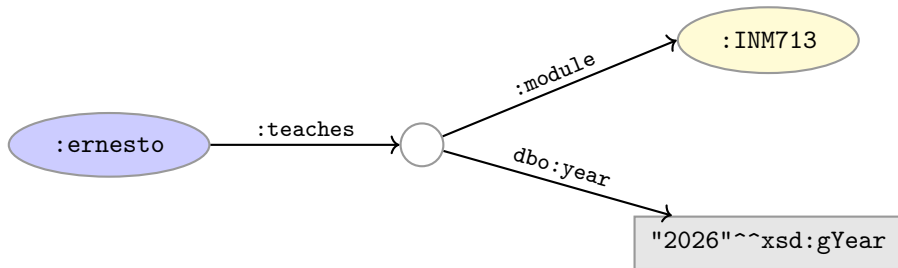
Annotating statements/triples in RDF (i)

RDF Solution 1: Named Graphs

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .  
@prefix dbo: <http://dbpedia.org/ontology/>  
@prefix : <http://www.example.org/university/london/city#> .  
  
:AcademicYear2026 {  
    :ernesto :teaches :INM713 .  
    :INM713 rdf:type :Module .  
    :INM713 :module_type "Elective"  
}  
:AcademicYear2026 dbo:year "2026"^^xsd:gYear .
```

Annotating statements/triples in RDF (ii)

RDF Solution 2: **n-ary relationships** (and blank-nodes)



Annotating statements/triples in RDF (iii)

RDF Solution 3: **Reification**

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .  
@prefix dbo: <http://dbpedia.org/ontology/>  
@prefix : <http://www.example.org/university/london/city#> .  
  
:ernesto :teaches :INM713 .  
_:s rdf:type rdf:Statement  
    rdf:subject :ernesto;  
    rdf:predicate :teaches ;  
    rdf:object :INM713 ;  
    dbo:year "2026"^^xsd:gYear .  
:aidan :likes _:s .
```

Annotating statements/triples in RDF (iv)

RDF Solution 4: **RDF-Star** now integrated within **RDF 1.2** (W3C draft)

- Uses triple operator « »
- Compact solution: less triples, less space, faster to load
- Closer to Property Graphs

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
```

```
@prefix dbo: <http://dbpedia.org/ontology/>
```

```
@prefix : <http://www.example.org/university/london/city#> .
```

```
:ernesto :teaches :INM713 .
```

```
<<:ernesto :teaches :INM713>> dbo:year "2026"^^xsd:gYear .
```

O. Hartig and B. Thompson. Foundations of an Alternative Approach to Reification in RDF. <https://arxiv.org/abs/1406.3399>.

Latest information: <https://w3c.github.io/rdf-star/>, <https://www.w3.org/TR/rdf12-primer/>

RDF and Property Graphs

- RDF workarounds/extensions to annotate triples.
- Is this a limitation?

RDF and Property Graphs

- RDF workarounds/extensions to annotate triples.
- Is this a limitation?
- More a **feature** due to its relation to **(logic-based) unary and binary predicates**.

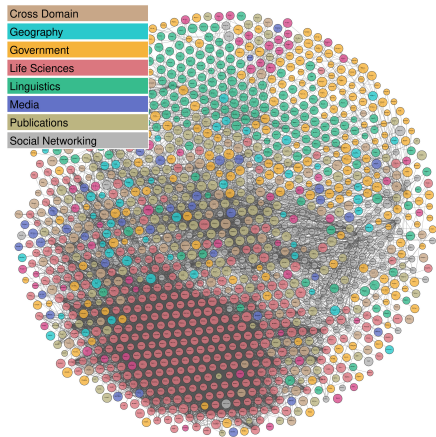
RDF and Property Graphs

- RDF workarounds/extensions to annotate triples.
- Is this a limitation?
- More a **feature** due to its relation to **(logic-based) unary and binary predicates**.
- **(OWL-layered) RDF-based KGs have clear semantics.**
- Property graphs do not have the same semantic commitment.

Knowledge Graphs on the Web and beyond

Open KGs

- Linked Open Datasets:
<https://lod-cloud.net/>
- Diverse domains: media, government, geography, tourism, life sciences, ecotoxicology, etc.
- General purpose KGs: DBpedia, Freebase, Wikidata, YAGO
 - <https://www.wikidata.org/>
 - <https://dbpedia.org/>



Domain specific KGs: Life Sciences

- **UMLS** (Unified Medical Language System) Metathesaurus
 - Integrates more than two hundred thesauri and ontologies
 - Contains more than 15 million concept names
- **BioPortal**
 - Contains more than 1,000 ontologies
 - Represent a network of ontologies
 - More than 79 million mappings are available
 - Also includes user-submitted alignments

Enterprise KGs (i)

- Websearch (*e.g.*, Bing, Google),
- Commerce (*e.g.*, Airbnb, Amazon, eBay, Uber),
- Social networks (*e.g.*, Facebook, LinkedIn),
- Finance (*e.g.*, Accenture, Banca d'Italia, Bloomberg, Capital One)

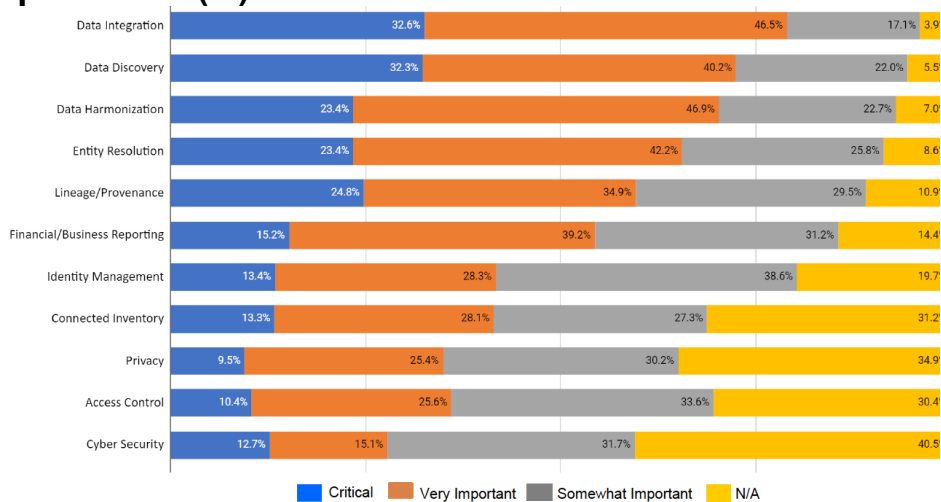
Aidan Hogan et al. Knowledge Graphs. Morgan & Claypool 2021.

Enterprise KGs (ii)

	Data model	Size of the graph	Development stage
Microsoft	The types of entities, relations, and attributes in the graph are defined in an ontology.	~2 billion primary entities, ~55 billion facts	Actively used in products
Google	Strongly typed entities, relations with domain and range inference	1 billion entities, 70 billion assertions	Actively used in products
Facebook	All of the attributes and relations are structured and strongly typed, and optionally indexed to enable efficient retrieval, search, and traversal.	~50 million primary entities, ~500 million assertions	Actively used in products
eBay	Entities and relation, well-structured and strongly typed	Expect around 100 million products, >1 billion triples	Early stages of development and deployment
IBM	Entities and relations with evidence information associated with them.	Various sizes. Proven on scales documents >100 million, relationships >5 billion, entities >100 million	Actively used in products and by clients

Natasha Noy et al. Industry-scale Knowledge Graphs: Lessons and Challenges. Communications of the ACM (2019).

Enterprise KGs (iii): Use cases



Enterprise Knowledge Graph Foundation: <https://www.ekgf.org>

How can we access the KGs?

- RDF dumps.
- From *SPARQL endpoints*:
 - Data kept in a *triple store*
 - Exposes data (in different formats)
 - with endpoint frontends, e.g.,
<http://dbpedia.org/resource/London>,
https://dbpedia.org/page/City,_University_of_London Or
 - by direct SPARQL query endpoint: e.g.,
<http://dbpedia.org/sparql>
<https://query.wikidata.org/>
<https://www.ebi.ac.uk/rdf/datasets/>

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- <https://www.uio.no/studier/emner/matnat/ifi/INF4580/>

Laboratory Session

Hands-on session on RDF

- Infrastructure set-up.
- Creating a small knowledge graph.
- First steps with Protégé.